

ELATMA, Russian Federation

WMO: 276480

Lat: 54.9722N

Lon: 41.7460E

Elev: 445

StdP: 14.46

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.0	-8.8	-18.7	2.0	-13.4	-13.5	2.7	-8.3	16.7	23.0	15.3	24.0	2.8	260	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	87.1	68.7	83.5	67.3	80.2	66.2	71.0	83.2	69.3	80.5	67.5	77.2	4.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.9	100.7	76.3	65.3	95.1	74.3	63.8	90.1	72.5	35.0	82.8	33.6	81.0	32.2	77.2	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	13.2	11.7	-21.3	91.3	6.6	3.7	-26.1	93.9	-29.9	96.0	-33.6	98.1	-38.4	100.8
			-21.3	73.9	6.3	2.1	-25.9	75.4	-29.6	76.6	-33.1	77.8	-37.7	79.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.9	16.0	17.5	27.3	43.5	56.5	62.8	67.7	63.9	53.5	41.5	29.8	20.7	(6)
	DBStd	20.49	12.66	12.46	8.54	9.27	8.89	7.30	6.46	7.16	7.28	7.87	9.44	11.88	(7)
	HDD50	4786	1054	909	703	240	38	2	0	45	281	605	909		(8)
	HDD65	8740	1519	1329	1168	646	289	127	46	108	351	728	1055	1374	(9)
	CDD50	1814	0	0	0	44	239	387	547	430	150	17	0	0	(10)
	CDD65	294	0	0	0	0	25	61	129	73	6	0	0	0	(11)
	CDH74	2658	0	0	0	3	256	546	1122	681	50	0	0	0	(12)
	CDH80	793	0	0	0	0	55	146	357	229	6	0	0	0	(13)
	WSAvg	5.3	5.9	5.9	5.9	5.5	5.2	4.8	4.0	4.3	4.5	5.6	6.0	6.1	(14)
	PrecAvg	23.8	1.7	1.5	1.1	1.4	1.6	2.2	3.5	2.5	2.1	2.2	2.2	1.9	(15)
(16) Precipitation	PrecMax	34.6	3.3	3.1	2.4	3.2	7.0	6.3	8.6	7.6	5.2	6.3	4.4	4.1	(16)
	PrecMin	14.9	0.0	0.0	0.4	0.2	0.0	0.2	0.4	0.0	0.2	0.0	0.4	0.1	(17)
	PrecStd	4.7	0.8	0.8	0.6	0.7	1.4	1.3	2.0	1.5	1.3	1.3	0.9	0.9	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.4	37.4	53.1	73.3	85.2	89.0	93.0	93.0	79.6	65.3	52.4	43.0	(19)
		MCWB	35.7	35.0	44.9	55.8	65.8	67.8	71.6	71.3	63.4	55.7	49.4	41.9	(20)
	2%	DB	34.9	35.6	45.8	68.2	80.7	83.7	87.3	86.1	73.8	59.5	47.3	39.0	(21)
		MCWB	33.5	33.7	39.0	53.7	63.2	66.5	68.9	69.6	61.6	53.8	45.3	37.7	(22)
	5%	DB	33.5	34.2	41.4	63.7	76.0	80.5	83.8	80.9	69.4	55.8	44.2	36.1	(23)
		MCWB	32.5	32.8	36.4	51.7	61.2	66.2	68.4	66.5	59.3	51.0	42.3	34.8	(24)
	10%	DB	32.2	33.0	38.1	59.3	71.9	76.5	80.4	76.8	65.4	52.7	41.0	33.9	(25)
		MCWB	31.4	31.9	34.6	49.1	58.7	64.4	67.5	64.5	57.4	49.2	39.7	32.9	(26)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.3	35.9	45.4	57.6	67.6	71.1	74.1	72.8	65.6	57.6	49.3	42.2	(27)
		MCDB	36.9	36.6	52.5	69.5	81.6	82.7	87.3	88.8	75.6	62.2	51.9	42.8	(28)
	2%	WB	33.9	34.2	39.8	55.2	64.6	69.2	71.6	70.2	63.1	54.5	45.8	38.0	(29)
		MCDB	34.6	34.8	44.4	65.8	77.2	80.5	83.4	83.9	71.4	58.6	47.0	38.8	(30)
	5%	WB	32.7	33.1	36.9	52.6	62.6	67.3	70.0	68.0	60.6	51.8	42.6	35.1	(31)
		MCDB	33.5	34.0	40.4	62.1	74.0	77.4	80.8	78.4	67.0	55.3	44.0	35.8	(32)
	10%	WB	31.3	32.0	35.1	50.0	60.4	65.4	68.3	65.8	58.4	49.5	39.7	33.1	(33)
		MCDB	32.2	33.2	37.6	58.1	70.5	74.3	77.7	74.5	64.4	52.2	40.9	33.7	(34)
(36) Mean Daily Temperature Range	5% DB	MDBR	10.2	12.0	13.7	17.1	19.3	17.8	18.5	18.4	16.0	11.0	7.8	8.7	(35)
		MCDBR	7.0	8.4	16.4	23.1	23.9	23.2	23.8	24.9	22.3	15.7	10.8	6.7	(36)
		MCWBR	7.1	7.5	10.7	11.9	11.0	9.9	9.7	10.3	11.6	10.5	9.1	6.4	(37)
	5% WB	MCDBR	7.0	7.9	13.9	21.1	22.1	20.5	20.3	22.0	19.4	14.0	9.7	6.7	(38)
		MCWBR	7.1	7.4	9.7	11.6	11.2	9.8	9.5	10.1	11.3	10.5	8.9	6.6	(39)
(40) Clear-Sky Solar Irradiance	taub		0.289	0.328	0.355	0.416	0.388	0.382	0.412	0.431	0.378	0.358	0.322	0.291	(40)
	taud		2.095	2.040	2.051	2.120	2.329	2.370	2.301	2.231	2.360	2.326	2.221	2.096	(41)
	Ebn at Noon		194	227	252	254	271	274	262	247	220	183	163		(42)
	Edh at Noon		21	32	40	43	37	36	38	38	30	24	19	17	(43)
(44) All-Sky Solar Radiation	RadAvg		214	503	999	1298	1750	1776	1787	1445	910	431	176	118	(44)
	RadStd		44	75	112	113	139	180	167	149	132	65	37	35	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	+1.33	+0.99	N/A	N/A	N/A	N/A	
(47)	Regional (6 neighbors)	+1.21	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page